

Course Code: SBT-DF204

Lab Title: Case Study 2 Reconstructing
Suspected Drug Trafficking From Chrome Web
History

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EXECUTIVE SUMMARY

This investigation examined a Google Chrome History SQLite database supplied as authorised course material. The purpose was to reconstruct browser activity associated with a suspected online drug-trafficking scenario.

The examination identified Craigslist posting activity, Gmail communication, a completed download named `proof_of_payment.png`, blockchain and Bitcoin-related browsing, and later Kraken activity on 19 April 2022.

The records form a connected sequence of browser activity. However, the Chrome History database alone does not prove who performed the activity, who owned the referenced accounts, or that an actual drug transaction occurred. The findings are therefore limited to the browser artefacts recorded in the supplied database.

LAB OBJECTIVE

The objective of this case study was to examine a Chrome browser History database and reconstruct browsing activities related to a suspected drug-trafficking investigation. The analysis focused on identifying relevant searches, Craigslist activity, downloads, cryptocurrency-related activity, and the timeline of events while preserving the integrity of the evidence.

TOOLS AND RESOURCES USED

- **Kali Linux:** Used as the forensic analysis environment.
- **SQLite3:** Used to examine the Chrome History database.
- **sha256sum:** Used to verify the SHA-256 integrity of the evidence.
- **cp --preserve=all:** Used to create preserved evidence and working copies.
- **ls -lh:** Used to verify the file size.

- **SQL queries:** Used to extract and correlate browser history records.
- **Chrome History database:** Used as the primary evidence.
- **Authorised course material:** Provided the History database used for the investigation.

All analysis was performed using the supplied training evidence. No live postings, accounts, exchanges, wallets, or transaction services were accessed.

EVIDENCE ACQUISITION AND INTEGRITY

Evidence Source

The evidence was a Google Chrome History SQLite database obtained from the authorised course material.

Filename: History

File size: 196,608 bytes (192 KB)

SHA-256: b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19

The supplied source was:

http://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Digital_Currency/LabFiles/History

Evidence Preservation

The original History file was preserved in the evidence folder, and a separate working copy was created for analysis. All SQL analysis was performed on the working copy to protect the original evidence.

Figure 1 — Original File Information

Hash Verification

The SHA-256 hash of both copies was checked.

Both copies produced:

b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19

The matching hashes confirm that the evidence copy and working copy were identical at the time of verification.

Figure 2 — Hash Verification

DATABASE EXAMINATION METHOD

Relevant Tables

The supplied file was examined as a SQLite database.

The main tables used were:

- urls
- visits
- downloads
- downloads_url_chains
- keyword_search_terms

The urls table provided URL and page-title information. The visits table provided visit IDs, timestamps, from_visit and transition values. The downloads tables provided information about downloaded files and their recorded source URLs. The keyword search table provided recorded search terms.

Figure 3 — Database Tables

Relevant Fields

The main fields examined were:

urls

- id
- url
- title
- visit_count
- typed_count
- last_visit_time

visits

- id
- url
- visit_time
- from_visit
- transition

downloads

- id
- guid
- current_path
- target_path
- start_time
- received_bytes
- total_bytes
- state
- mime_type
- site_url

downloads_url_chains

- id
- chain_index
- url

Figure 4 — Database Schema

Timestamp Conversion

Chrome timestamps are stored as microseconds from 1 January 1601 UTC. The timestamps were converted to readable UTC dates and times using SQLite date conversion. The same method was applied to the download timestamps.

Transition Analysis

The browser transition value was examined to understand how pages were accessed. The relevant values were 0 for normal navigation, 5 for omnibox-generated navigation, and 7 for form submission.

The Craigslist posting records showed a transition value of 7, supporting the recorded posting workflow. The from_visit field was also used to link related browser visits.

FINDINGS AND ANALYSIS

Question 1: What evidence file was acquired and how was it preserved?

The evidence was a Chrome History SQLite database named History, supplied through the authorised course material.

The file was preserved by creating a separate evidence copy and a separate working copy. SHA-256 verification produced the same hash for both copies:

b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19

Analysis was performed only on the working copy.

Finding: The evidence copies were verified as identical at the time of examination.

Question 2: What browser-history tables and fields were relevant?

The main tables were urls, visits, downloads, downloads_url_chains and keyword_search_terms.

The urls and visits tables were joined using:

```
visits.url = urls.id
```

This linked each browser visit to its URL and page title.

The visits.from_visit and visits.transition fields were used to reconstruct navigation relationships.

Finding: These tables provided the information required to reconstruct the relevant browsing sequence.

Question 3: When did the recorded activity occur?

The relevant activity occurred on 19 April 2022.

The first relevant browser visit identified was:

13:54:18 UTC — Visit ID 1

The later relevant Gmail record was:

15:02:56 UTC — Visit ID 74

The activity therefore covers approximately 68 minutes and 38 seconds.

All report times are presented in UTC.

Finding: The relevant browser activity occurred within a continuous recorded period on 19 April 2022.

Question 4: How did the browser reach the relevant URLs?

The from_visit and transition fields were used to understand the navigation sequence.

The Craigslist posting workflow included transition type 7, which corresponds to form submission. The Google searches for Craigslist, blockchain explorer, and Kraken recorded transition type 5, corresponding to omnibox-generated navigation.

The from_visit values also connect the successive Craigslist pages, showing the progression through category selection, posting creation, map activity, image editing, preview, and confirmation.

Finding: The transition and from_visit records support the reconstructed navigation sequence.

Question 5: What Craigslist activity occurred?

At 13:54:18 UTC, the browser recorded a Google search for:

craigslist

This was followed by Craigslist account and login activity.

At approximately 13:55:05 UTC, the browser entered the Craigslist posting workflow. The recorded pages included:

- Choose type
- Choose category
- Create posting
- Add map
- Edit image
- Preview
- Posting confirmation
- Manage posting
- Listing

At 13:58:31 UTC, Visit ID 21 accessed URL ID 16.

The listing title was:

“cheaper than Rx supplements - health and beauty - by owner -...”

Finding: The Chrome History records a sequence consistent with creating and managing a Craigslist listing.

The History database does not contain the complete submitted form contents and does not identify the person who performed the activity.

Figure 5 — Craigslist Posting Timeline

Question 6: What communication and payment-related activity occurred?

At 14:04:12 UTC, Visit ID 43 accessed a Gmail page titled:

“cheaper than Rx supplements - order - unsub.fscs@gmail.com - Gmail”

Additional Gmail activity followed.

At 14:56:47 UTC, Visit ID 48 accessed another Gmail page containing the transaction ID:

517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5

At 14:56:58 UTC, the downloads table recorded a file named:

proof_of_payment.png

- Download ID: 1
- Size: 33,844 bytes
- MIME type: image/png
- State: 1
- Recorded site: Imgur

The download URL chain recorded the corresponding image URL on Imgur.

Finding: Chrome recorded Gmail activity related to the listing and transaction ID, followed by a PNG download named proof_of_payment.png.

The filename suggests a payment-related image, but the History database does not establish the contents or authenticity of the image.

Figure 6 — Relevant Browser Timeline

Insert Screenshot_05_Relevant_Browser_Timeline.png

Figure 7 — Download Evidence

Insert Screenshot_06_Download_Evidence.png

Figure 8 — Download URL Chain

Insert Screenshot_07_Download_URL_Chain.png

Question 7: What cryptocurrency-related activity occurred?

At 14:58:59 UTC, Visit ID 51 recorded a Google search for “blockchain explorer.” This was followed by access to a blockchain explorer.

At 14:59:06 UTC, the browser accessed a transaction search containing the recorded transaction ID and then a Bitcoin transaction page.

At 14:59:21 UTC, Visit ID 58 accessed a Bitcoin address page containing:

38RcsURWCDmCbYocmsZCnGz1FtD7x477mt

At 14:59:38 UTC, the browser recorded a Google search for “Kraken.”

This was followed by access to several Kraken pages, including:

- Sign-in
- Device approval
- Trade
- Instant
- Funding
- Transaction-related redirects
- Ledger

The funding page was titled:

“41,543.80 USD - Kraken - Funding - Overview”

The later ledger page was titled:

“41,563.70 USD - Kraken - History - Ledger”

Finding: The browser history records cryptocurrency-related browsing, including Bitcoin transaction and address pages and Kraken account-related pages.

The History database alone does not establish account ownership or prove that a financial transaction was completed.

Figure 9 — Relevant Browser Timeline

Insert Screenshot_05_Relevant_Browser_Timeline.png

Question 8: How do the events correlate?

The recorded activity forms the following sequence:

Craigslist search → Craigslist posting → Gmail order activity → transaction-ID-related Gmail activity → payment-related download → blockchain/Bitcoin activity → Kraken activity → return to Gmail

The close timing of these records provides a basis for correlating them as part of the same browser activity sequence.

However, correlation alone does not prove that the activity represented an actual drug transaction.

Finding: The browser records provide a consistent chronological sequence, but the History database alone cannot establish the complete context or intent behind the activity.

RECONSTRUCTED TIMELINE

1. 13:54:18 UTC

Visit 1 / URL 1 — Google search for Craigslist.

2. 13:54:19 UTC

Visit 3 / URL 2 — Craigslist homepage.

3. 13:54:22 UTC

Visits 5–7 — Craigslist account and login activity.

4. 13:55:05 UTC

Visit 11 / URL 7 — Craigslist posting workflow begins.

5. 13:55:10 UTC

Visit 14 / URL 10 — Choose category.

6. 13:55:15 UTC

Visit 15 / URL 11 — Create posting.

7. 13:58:13 UTC

Visit 16 / URL 12 — Add map.

8. 13:58:17 UTC

Visit 17 / URL 13 — Edit image.

9. 13:58:19 UTC

Visit 18 / URL 14 — Preview.

10. 13:58:22 UTC

Visit 19 / URL 8 — Posting confirmation.

11. 13:58:31 UTC

Visit 21 / URL 16 — Listing titled **“cheaper than Rx supplements.”**

12. 14:04:12 UTC

Visit 43 / URL 34 — Gmail order-related message.

13. 14:56:47 UTC

Visit 48 / URL 37 — Gmail transaction-ID-related message.

14. 14:56:58 UTC

Download 1 — proof_of_payment.png recorded as downloaded.

15. 14:58:59 UTC

Visit 51 / URL 40 — Search for “blockchain explorer.”

16. 14:59:06 UTC

Visits 55–57 — Bitcoin transaction-related pages.

17. 14:59:21 UTC

Visit 58 / URL 44 — Bitcoin address page.

18. 14:59:38 UTC

Visit 59 / URL 45 — Search for “Kraken.”

19. 14:59:47 UTC

Visits 61–62 — Kraken sign-in activity.

20. 15:00:21 UTC

Visit 63 / URL 47 — Kraken device approval.

21. 15:01:11–15:01:21 UTC

Visits 64–66 — Kraken trade, instant, and funding pages.

22. 15:01:31–15:01:33 UTC

Visits 67–70 — Kraken/Mempool transaction-related navigation.

23. 15:02:05 UTC

Visit 71 / URL 55 — Kraken ledger accessed.

24. 15:02:56 UTC

Visit 74 / URL 37 — Gmail transaction-related message accessed again.

CONCLUSION

The Chrome History examination reconstructed browser activity on 19 April 2022, including a Craigslist posting titled “cheaper than Rx supplements,” related Gmail activity, a proof_of_payment.png download, and subsequent Bitcoin and Kraken-related browsing.

The timestamps, URLs, navigation records, download records, and search terms support the reconstructed sequence.

The evidence establishes the recorded browser activities and their sequence. However, additional forensic evidence would be required to confirm the identity of the user, ownership of the accounts, authenticity of the downloaded image, and whether an actual drug transaction took place.

Overall, the findings provide a consistent record of the investigated browser activity while avoiding conclusions that cannot be established from the Chrome History database alone.

EVIDENCE LOG

E01 — Craigslist Posting Activity

Source: Visits 11–21 / URLs 7–16

Finding: Craigslist posting workflow and listing titled “cheaper than Rx supplements.”

Reference: Figure 5

E02 — Gmail Activity

Source: Visits 43–48 / URLs 34–37

Finding: Gmail order and transaction-related activity.

Reference: Figure 6

E03 — Download Evidence

Source: Download ID 1

Finding: proof_of_payment.png recorded as downloaded.

Reference: Figure 7

E04 — Download URL Chain

Source: Download chain ID 1

Finding: Imgur image URL recorded.

Reference: Figure 8

E05 — Cryptocurrency Activity

Source: Visits 51–58 / URLs 40–44

Finding: Blockchain explorer, Bitcoin transaction, and Bitcoin address activity.

Reference: Figure 6

E06 — Recorded Search Terms

Source: Keyword search records

Finding: Searches for Craigslist, Gmail, blockchain explorer, and Kraken.

Reference: Figure 9

E07 — Kraken Activity

Source: Visits 59–71 / URLs 45–55

Finding: Kraken sign-in, device approval, trade, funding, transaction-related navigation, and ledger activity.

Reference: Figure 6

E08 — Craigslist Transition Records

Source: Visits 14–19

Finding: Raw transition value 805306375, with transition type 7, supporting form-submission navigation during the Craigslist posting workflow.

Reference: Question 4

E09 — Search Transition Records

Source: Visits 51 and 59

Finding: Raw transition value 838860805, with transition type 5, supporting omnibox-generated navigation for the blockchain explorer and Kraken searches.

Reference: Question 4

E10 — Evidence Integrity

Source: Evidence and working copies

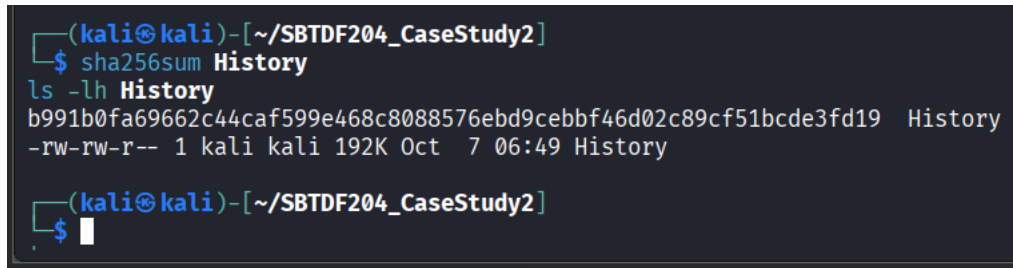
Finding: Matching SHA-256 hashes confirmed that the evidence and working copies were identical.

Reference: Figure 2

APPENDICES

Appendix A — Evidence Screenshots

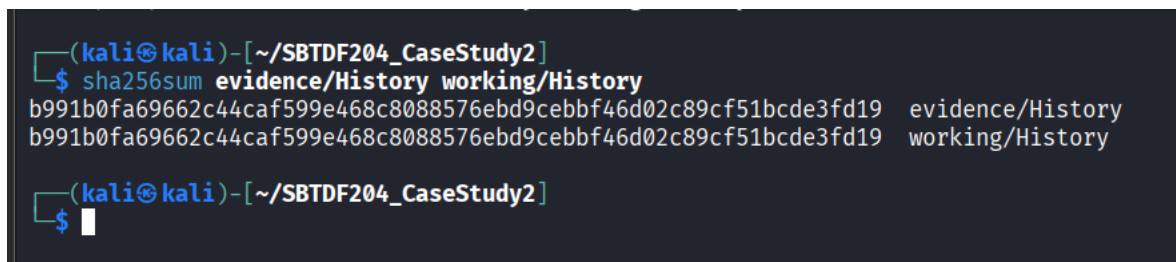
Figure 1 — Original File Information

A terminal window with a dark background. The prompt is (kali@kali)~[~/SBTDF204_CaseStudy2]. The user enters 'sha256sum History' and 'ls -lh History'. The output shows the SHA256 hash and file details for 'History'.

```
(kali@kali)~[~/SBTDF204_CaseStudy2]
$ sha256sum History
b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19 History
$ ls -lh History
-rw-rw-r-- 1 kali kali 192K Oct  7 06:49 History
(kali@kali)~[~/SBTDF204_CaseStudy2]
$
```

Screenshot_01_Original_File_Information.png

Figure 2 — Hash Verification

A terminal window with a dark background. The prompt is (kali@kali)~[~/SBTDF204_CaseStudy2]. The user enters 'sha256sum evidence/History working/History'. The output shows the SHA256 hash for both files, which match.

```
(kali@kali)~[~/SBTDF204_CaseStudy2]
$ sha256sum evidence/History working/History
b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19 evidence/History
b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19 working/History
(kali@kali)~[~/SBTDF204_CaseStudy2]
$
```

Screenshot_02_Hash_Verification.png

Figure 3 — Database Tables

```
(kali㉿kali)-[~/SBTDF204_CaseStudy2]
$ sqlite3 working/History "SELECT name FROM sqlite_master WHERE type='table' ORDER BY name;"
clusters
clusters_and_visits
content_annotations
context_annotations
downloads
downloads_reroute_info
downloads_slices
downloads_url_chains
keyword_search_terms
meta
segment_usage
segments
sqlite_sequence
typed_url_sync_metadata
urls
visit_source
visits

(kali㉿kali)-[~/SBTDF204_CaseStudy2]
$
```

Screenshot_03_Database_Tables.png

Figure 4 — Database Schema

```
(kali㉿kali)-[~/SBTDF204_CaseStudy2]
$ sqlite3 -header -column working/History "PRAGMA table_info(urls); PRAGMA table_info(visits); PRAGMA table_info(downloads); PRAGMA table_info(downloads_u"
cid  name                type      notnull  dflt_value  pk
--  --
0    id                  INTEGER   0        0           1
1    url                 LONGVARCHAR 0        0           0
2    title              LONGVARCHAR 0        0           0
3    visit_count        INTEGER    1        0           0
4    typed_count        INTEGER    1        0           0
5    last_visit_time    INTEGER    1        0           0
6    hidden             INTEGER    1        0           0
cid  name                type      notnull  dflt_value  pk
--  --
0    id                  INTEGER   0        0           1
1    url                 INTEGER    1        0           0
2    visit_time          INTEGER    1        0           0
3    from_visit          INTEGER    0        0           0
4    transition          INTEGER    1        0           0
5    segment_id          INTEGER    0        0           0
6    visit_duration      INTEGER    1        0           0
7    incremented_omnibox_typed_score  BOOLEAN   1        FALSE        0
8    opener_visit        INTEGER    0        0           0
cid  name                type      notnull  dflt_value  pk
--  --
0    id                  INTEGER   0        0           1
1    guid               VARCHAR    1        0           0
2    current_path        LONGVARCHAR 1        0           0
3    target_path         LONGVARCHAR 1        0           0
4    start_time          INTEGER    1        0           0
5    received_bytes      INTEGER    1        0           0
6    total_bytes         INTEGER    1        0           0
7    state               INTEGER    1        0           0
```

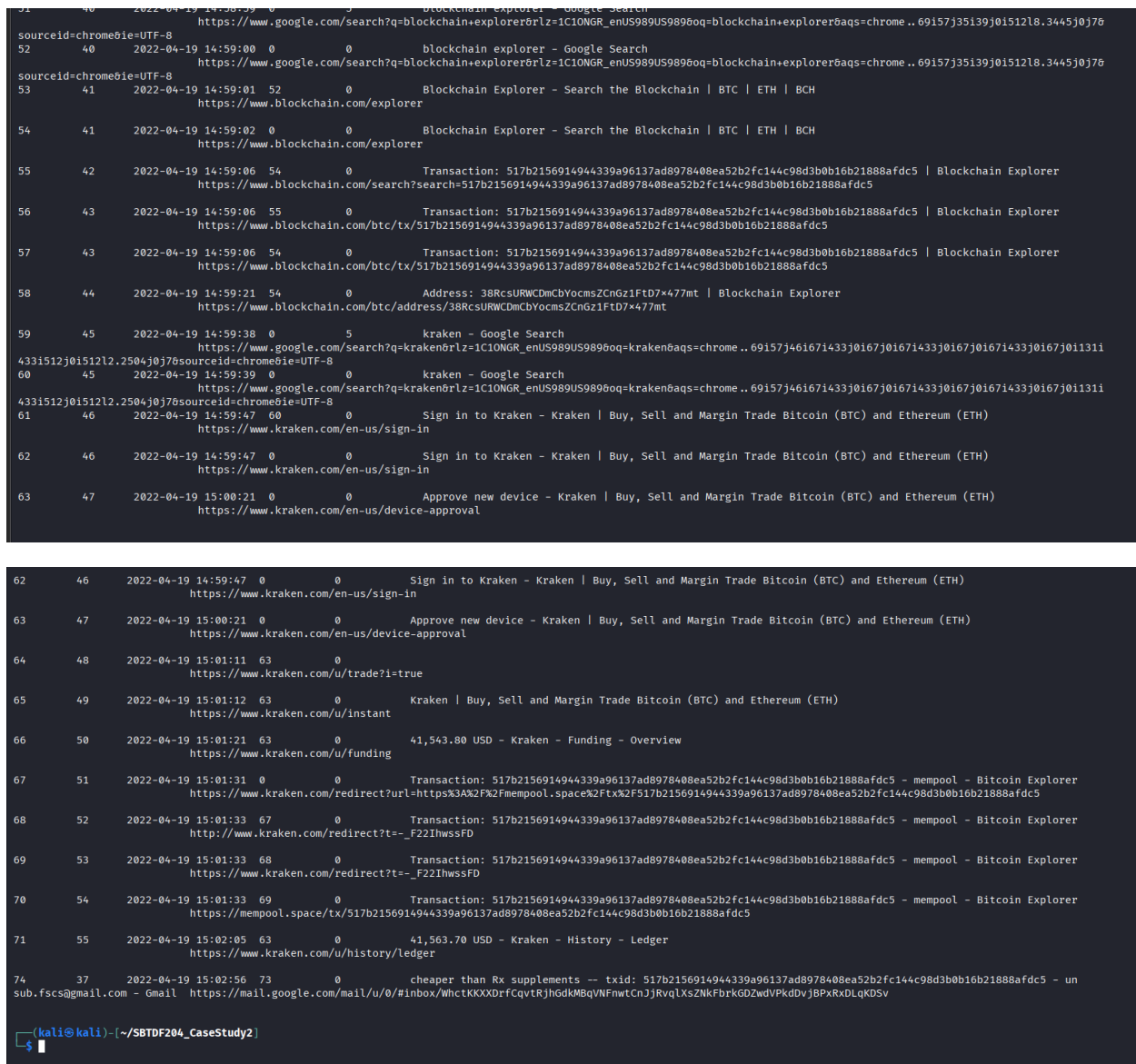
5	received_bytes	INTEGER	1	0	
6	total_bytes	INTEGER	1	0	
7	state	INTEGER	1	0	
8	danger_type	INTEGER	1	0	
9	interrupt_reason	INTEGER	1	0	
10	hash	BLOB	1	0	
11	end_time	INTEGER	1	0	
12	opened	INTEGER	1	0	
13	last_access_time	INTEGER	1	0	
14	transient	INTEGER	1	0	
15	referrer	VARCHAR	1	0	
16	site_url	VARCHAR	1	0	
17	tab_url	VARCHAR	1	0	
18	tab_referrer_url	VARCHAR	1	0	
19	http_method	VARCHAR	1	0	
20	by_ext_id	VARCHAR	1	0	
21	by_ext_name	VARCHAR	1	0	
22	etag	VARCHAR	1	0	
23	last_modified	VARCHAR	1	0	
24	mime_type	VARCHAR(255)	1	0	
25	original_mime_type	VARCHAR(255)	1	0	
26	embedder_download_data	VARCHAR	1	''	0
cid	name	type	notnull	dflt_value	pk
0	id	INTEGER	1		1
1	chain_index	INTEGER	1		2
2	url	LONGVARCHAR	1		0

Screenshot_04_Database_Schema.png

Figure 5 — Relevant Browser Timeline

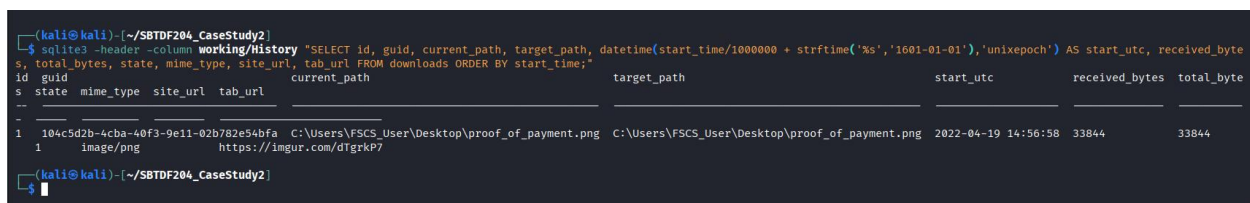
```
(kali@kali)-[~/SBTDF204_CaseStudy2]
$ sqlite3 -header -column working/History "SELECT v.id AS visit_id, v.url AS url_id, datetime(v.visit_time/1000000 + strftime('%s','1601-01-01'),'unixepoch') AS visit_
utc, v.from_visit, (v.transition & 255) AS transition, u.title, u.url FROM visits v JOIN urls u ON v.url=u.id WHERE v.url BETWEEN 34 AND 55 ORDER BY v.visit_time;"
visit_id url_id visit_utc from_visit transition title
url
```

43	34	2022-04-19 14:04:12	0	0	cheaper than Rx supplements - order - unsub.fscs@gmail.com - Gmail
					https://mail.google.com/mail/u/0/#inbox/whctKXXXXrFqCqXBGCnHkHGxZnZbkXXdXnJtmxwZwPtjVmvHMrFPPrthPXJgXngHcFLmggWv
44	35	2022-04-19 14:04:35	43	0	cheaper than Rx supplements - order - unsub.fscs@gmail.com - Gmail
					https://mail.google.com/mail/u/0/#inbox/whctKXXXXrFqCqXBGCnHkHGxZnZbkXXdXnJtmxwZwPtjVmvHMrFPPrthPXJgXngHcFLmggWv?compose=JHrtffdgkFjdrvzMFJPRM
45	36	2022-04-19 14:04:46	44	0	cheaper than Rx supplements - order - unsub.fscs@gmail.com - Gmail
					https://mail.google.com/mail/u/0/#inbox/whctKXXXXrFqCqXBGCnHkHGxZnZbkXXdXnJtmxwZwPtjVmvHMrFPPrthPXJgXngHcFLmggWv?compose=DXDwSWwqgZwzDRTRJhRT
46	34	2022-04-19 14:04:52	45	0	cheaper than Rx supplements - order - unsub.fscs@gmail.com - Gmail
					https://mail.google.com/mail/u/0/#inbox/whctKXXXXrFqCqXBGCnHkHGxZnZbkXXdXnJtmxwZwPtjVmvHMrFPPrthPXJgXngHcFLmggWv
48	37	2022-04-19 14:56:47	0	0	cheaper than Rx supplements -- txid: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 - un
					sub.fscs@gmail.com - Gmail https://mail.google.com/mail/u/0/#inbox/whctKXXXXrFqCqXBGCnHkHGxZnZbkXXdXnJtmxwZwPtjVmvHMrFPPrthPXJgXngHcFLmggWv
49	38	2022-04-19 14:56:52	0	0	Imgur: The magic of the Internet
					https://www.google.com/url?q=https://imgur.com/dTgrkP7&source=gmail&ust=1650466604648000&usg=AOvVaw0058uwJx6Jub4I72ZRaQpM
50	39	2022-04-19 14:56:52	49	0	Imgur: The magic of the Internet
					https://imgur.com/dTgrkP7
51	40	2022-04-19 14:58:59	0	5	blockchain explorer - Google Search
					https://www.google.com/search?q=blockchain+explorer&rlz=1C10NGR_enUS989US9896oq=blockchain+explorer&aqs=chrome..69157j35139j0i512l8.3445j0j7&
52	40	2022-04-19 14:59:00	0	0	blockchain explorer - Google Search
					https://www.google.com/search?q=blockchain+explorer&rlz=1C10NGR_enUS989US9896oq=blockchain+explorer&aqs=chrome..69157j35139j0i512l8.3445j0j7&
53	41	2022-04-19 14:59:01	52	0	Blockchain Explorer - Search the Blockchain BTC ETH BCH
					https://www.blockchain.com/explorer



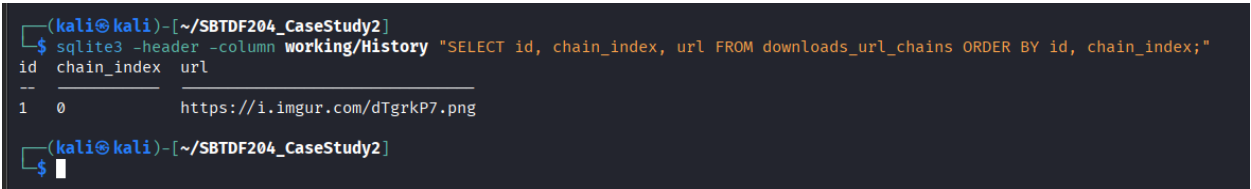
Screenshot_05_Relevant_Browser_Timeline.png

Figure 6 — Download Evidence



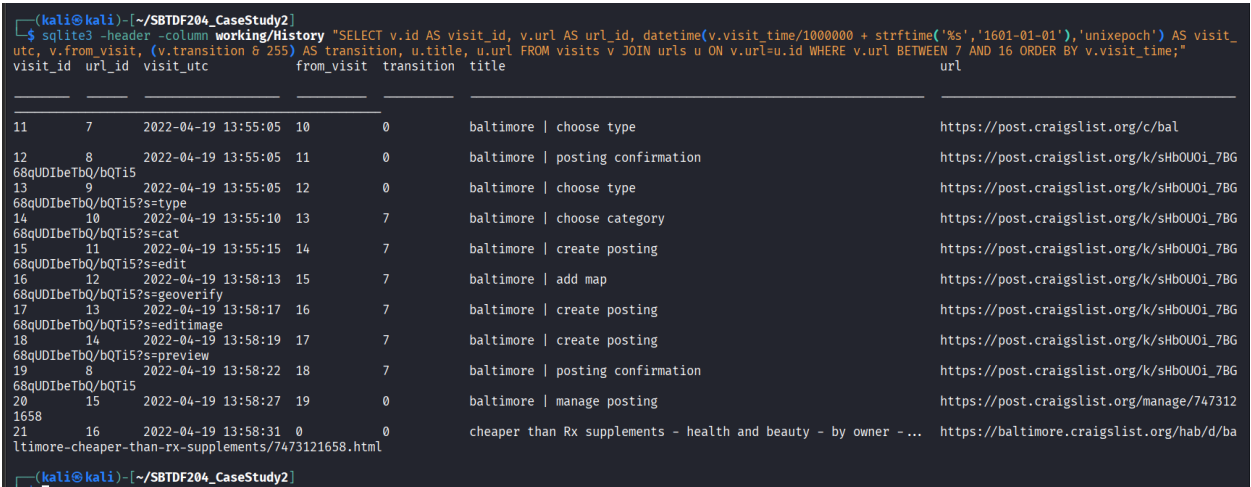
Screenshot_06_Download_Evidence.png

Figure 7 — Download URL Chain



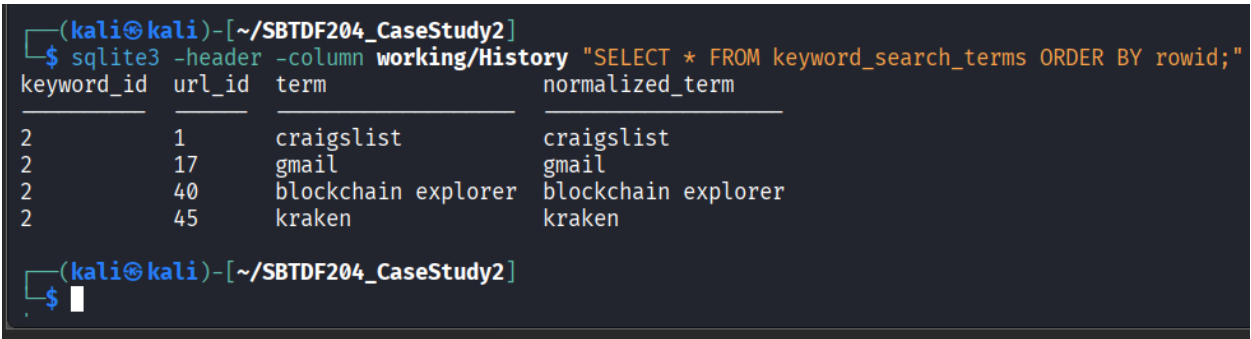
Screenshot_07_Download_URL_Chain.png

Figure 8 — Craigslist Posting Timeline



Screenshot_08_Craigslist_Posting_Timeline.png

Figure 9 — Recorded Search Terms



Screenshot_09_Recorded_Search_Terms.png

Appendix B— Evidence Integrity

Evidence SHA-256:

b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19

Working Copy SHA-256:

b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19

The matching SHA-256 hashes confirm that the preserved evidence copy and working copy were identical at the time of verification.